

HIGHER-MOMENT POLE LADDERS IN PRODUCT-PLUS-ONE COALESCENCE: A YULE-AVERAGED MARKED REFINEMENT

ANONYMOUS

ABSTRACT. Start with n ordered copies of 1, repeatedly choose a current adjacent pair uniformly, and replace (x, y) by $xy + 1$; write X_n for the terminal value. Disanto, Fuchs, Paningbatan, and Rosenberg proved that the number R_n of root ancestral configurations in a Yule–Harding random tree obeys the same recursion after the shift $X_n = R_n + 1$. We prove the identification under the common ordered-history coupling and assign their split law, finite distribution, unmarked antichain interpretation, mean analysis, and second-moment neighborhood zero contribution credit. After this subtraction, and after crediting the existing fixed-tree cardinality marker, we obtain its Yule-averaged bivariate transform in closed form. The all-order moment equation is a mechanical expansion of the owned exact law. The main residual theorem continues the two owned low-order radii to a strict ladder: for every $r \geq 3$, the r th raw-moment series has a normalized positive simple pole at a radius $\rho_r < \rho_{r-1}$ and $\limsup_n (\mathbb{E}X_n^r)^{1/n} = \rho_r^{-1}$. No full coefficient asymptotic is claimed for $r \geq 3$. The residual has passed only a bounded owner audit; novelty, priority, and external release remain open.

1. INTRODUCTION AND SCOPE

Consider the following rank-decreasing random dynamics. Begin with the ordered word $(1, \dots, 1)$ of length n . If the current word has j entries, choose one of its $j - 1$ adjacent pairs with probability $1/(j - 1)$ and apply

$$(1) \quad (x, y) \mapsto xy + 1.$$

After $n - 1$ mergers a positive integer X_n remains. The local rule is nonassociative, so X_n depends on the entire merger genealogy. The first nondegenerate law already occurs at $n = 4$:

$$\mathbb{P}(X_4 = 4) = \frac{2}{3}, \quad \mathbb{P}(X_4 = 5) = \frac{1}{3}.$$

The decisive owner is not merely a generic random-tree neighbor. Disanto et al. [3] study the number R_n of root ancestral configurations under the Yule–Harding model. Their Proposition 3.5 gives $R_1 = 0$ and the uniform-split recurrence $R_n \stackrel{d}{=} (R_I + 1)(R'_{n-I} + 1)$, including the exact finite law. Their Section 3.2 identifies root configurations with nonempty antichains of the pruned internal-node tree; Sections 5.2–5.3 derive the mean and second-moment Riccati analyses. Thus $X_n = R_n + 1$ is the same statistic in shifted notation. All of those statements, including the mean pole $1 - \exp(-2\pi/(3\sqrt{3}))$ and the order-two growth analysis, receive zero contribution credit here.

Two further owners narrow the residual. The cardinality refinement on a fixed rooted-tree poset is studied by Andriantiana et al. [1]; only its average under the Yule/uniform ordered-history law and the resulting closed bivariate transform are at issue here. Moreover, Chang and Fuchs [2, Table 1] give the exact Yule–Harding probability $2^{n-2}/(n - 1)!$ of an n -caterpillar, building on the caterpillar analysis of Rosenberg [10]. Both the minimizing shape and its probability therefore receive zero credit.

The genealogy itself is also classical. Each current boundary is an undeleted original boundary, and successive uniform choices form a uniform permutation. Its max-Cartesian tree has the usual random binary-search-tree (BST) root-split law. The same subtraction applies to deterministic subtree enumeration [11], ideals of forest posets [8], antichain statistics on rooted plane trees [7], and random forest ideals [6]. Random-BST generating-function schemes and singularity methods are also established [4, 5, 9].

After those deductions, the paper records only the following residual package:

- (i) the Yule-averaged bivariate ordinary generating function for the owned fixed-tree cardinality marker, together with its closed Euler form; and
- (ii) beginning at $r = 3$, a strict continuation of the moment-radius ladder with unit residues and exact exponential limsup. The all-order triangular identity is retained only as the mechanical interface from the owned exact law.

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The higher-moment conclusion is deliberately limited to a positive simple pole and an exact exponential *limsup*. Without control of all complex singularities, no full asymptotic is asserted for $r \geq 3$. The direct owner has a stronger order-two conclusion than the ceiling used here.

Two complementary viewpoints organize the proofs. The temporal viewpoint uses uniform deletion orders, conditional root splits, moments, and Riccati–Euler equations. The structural viewpoint regards the same orders as heap labelings of Cartesian trees and interprets the output as a cardinality-marked antichain count. Both viewpoints use the same split law; we do not present them as probabilistically independent proofs. The second viewpoint supplies a Yule-averaged cardinality transform and exact comb controls rather than a new random-tree model. A focused search of the direct owner’s citation neighborhood and current root-configuration formulations did not locate the averaged transform or a strict $r \geq 3$ continuation of the owned low-order pole ladder. This bounded non-hit is not evidence of novelty or priority. External circulation remains HOLD.

2. DIRECT-OWNER INTERFACE AND THE EXACT SHIFT

Label the $n - 1$ original boundaries by $1, \dots, n - 1$. A merger deletes exactly the boundary separating the two merged blocks; every other current boundary retains its original label.

Theorem 2.1 (Owned split law and objectwise shift). *Let I_n be uniform on $\{1, \dots, n - 1\}$. Conditional on $I_n = i$, let X_i and X'_{n-i} be independent copies generated on the two sides of the last-deleted boundary. Then*

$$(2) \quad X_1 = 1, \quad X_n \stackrel{d}{=} 1 + X_{I_n} X'_{n-I_n} \quad (n \geq 2).$$

Consequently, if $p_n(v) = \mathbb{P}(X_n = v)$, then $p_1(v) = \mathbf{1}_{\{v=1\}}$ and

$$(3) \quad p_n(v) = \frac{1}{n-1} \sum_{i=1}^{n-1} \sum_{\substack{a,b \geq 1 \\ 1+ab=v}} p_i(a) p_{n-i}(b).$$

Every law in (3) has finite support. Moreover, couple the merger process to the ordered unlabeled history having the same boundary order. If R_n counts its root configurations, then

$$(4) \quad X_n = R_n + 1$$

for every history, not only in distribution.

Proof. When j blocks remain, precisely $j - 1$ original boundaries remain, and the process selects each with probability $1/(j - 1)$. Thus the complete deletion order is a uniform permutation of $\{1, \dots, n - 1\}$. Its last element is uniform. Conditional on that element being i , the relative orders of the $i - 1$ left boundaries and the $n - i - 1$ right boundaries are independent and uniform. All mergers on each side occur before the last deletion, after which (1) combines the two terminal side values. This proves (2). Conditioning on I_n and the two side values gives (3). Finiteness follows inductively. These distributional statements reproduce Proposition 3.5 of Disanto et al. [3] after a shift and receive zero credit.

For the objectwise identification, let T be the common ordered evaluation tree. At a leaf, its root-configuration count is $R(T) = 0$, while the merger value is $V(T) = 1$. At an internal root, the standard root-configuration decomposition gives

$$R(T) = (R(T_L) + 1)(R(T_R) + 1).$$

Induction and $V(T) = 1 + V(T_L)V(T_R)$ now give $V(T) = R(T) + 1$. This proves (4); the root-configuration recurrence and its unmarked antichain interpretation are owned by Disanto et al. [3]. $\square$

The proof also fixes the orientation: the *last* deleted boundary is the root of the evaluation tree. Reversing the deletion order exchanges maxima and minima in the Cartesian-tree description and is not the process above.

3. HEAP LABELINGS AND A MARKED ANTICHAIN TRANSFORM

For a fixed deletion order, let T be the ordered full binary evaluation tree. Its leaves carry 1, and every internal vertex applies $V(T) = 1 + V(T_L)V(T_R)$. Order the internal vertices by ancestry. For an indeterminate s , define

$$P_T(s) = \sum_B s^{|B|},$$

where B ranges over antichains of internal vertices, including the empty antichain. Put $P_{\text{leaf}}(s) = 1$.

Lemma 3.1 (Owned unmarked interface and its marker). *For every evaluation tree,*

$$(5) \quad P_T(s) = s + P_{T_L}(s)P_{T_R}(s), \quad V(T) = P_T(1).$$

Proof. An antichain either contains the root, in which case it is the singleton root, or excludes the root, in which case its restrictions to the two subtrees are arbitrary antichains and can be chosen independently. The first identity follows. Its specialization at $s = 1$ is the same recursion and initial condition as $V(T)$. $\square$

At $s = 1$, this is precisely the direct owner's correspondence: $R(T)$ counts nonempty antichains and $X(T) = R(T) + 1$ also includes the empty one. For general s , the fixed-tree cardinality marker and the equivalent root-subtree leaf statistic are also owned by Andriantiana et al. [1]. Only averaging this marker under the Yule/uniform ordered-history law and solving the resulting bivariate equation belong to the residual claim set.

Let T_n denote the tree induced by a uniform boundary order and write

$$a_n(s) = \mathbb{E}P_{T_n}(s), \quad A(z, s) = \sum_{n \geq 1} a_n(s)z^{n-1}.$$

Theorem 3.2 (Expected cardinality-marked antichains). *Coefficientwise as a formal series in z and s ,*

$$(6) \quad \partial_z A = A^2 + \frac{s}{(1-z)^2}, \quad A(0, s) = 1.$$

Set $w = 1 - z$,

$$\delta = \sqrt{1 - 4s}, \quad \beta_{\pm} = \frac{1 \pm \delta}{2},$$

and

$$(7) \quad Y(w, s) = \frac{\beta_+ w^{\beta_+} - \beta_- w^{\beta_-}}{\delta}.$$

With the removable value

$$Y(w, 1/4) = w^{1/2} \left(1 + \frac{1}{2} \log w \right),$$

the solution is

$$(8) \quad A(z, s) = \frac{Y_w(1 - z, s)}{Y(1 - z, s)}.$$

In particular, $[z^{n-1}s^k]A$ is the exact expected number of k -element internal-node antichains, and $A(z, 1)$ is the mean series of X_n .

Proof. The split law and lemma 3.1 give, for $n \geq 2$,

$$(9) \quad a_n(s) = s + \frac{1}{n-1} \sum_{i=1}^{n-1} a_i(s)a_{n-i}(s).$$

Multiplying by $(n-1)z^{n-2}$ and summing gives (6). The function in (7) satisfies

$$Y_{ww} + \frac{s}{w^2}Y = 0, \quad Y(1, s) = Y_w(1, s) = 1.$$

Because $\partial_z = -\partial_w$, the logarithmic derivative $A = Y_w/Y$ satisfies (6) and the initial condition. Uniqueness of formal solutions proves (8). Taking the limit $\delta \rightarrow 0$ in (7) gives the displayed removable value. The coefficient interpretation is the definition of $a_n(s)$, and lemma 3.1 gives $a_n(1) = \mathbb{E}X_n$. $\square$

Equation (8) is interpreted formally near $s = 0$, or analytically with consistent branches. At $s = 0$ it gives $A(z, 0) = (1 - z)^{-1}$, recording the unique empty antichain. The marked transform adds a coefficient-level statement, but its derivation still uses the same classical root split as theorem 2.1.

4. ALL RAW MOMENTS AND A STRICT RADIUS CASCADE

For integers $r \geq 0$ set

$$m_{r,n} = \mathbb{E}X_n^r, \quad F_r(z) = \sum_{n \geq 1} m_{r,n}z^{n-1}.$$

Thus $m_{0,n} = 1$ and $F_0(z) = (1 - z)^{-1}$.

Proposition 4.1 (Mechanical all-order moment interface). *For $n \geq 2$ and $r \geq 0$,*

$$(10) \quad (n-1)m_{r,n} = \sum_{i=1}^{n-1} \sum_{k=0}^r \binom{r}{k} m_{k,i} m_{k,n-i}.$$

Equivalently,

$$(11) \quad F'_r(z) = \sum_{k=0}^r \binom{r}{k} F_k(z)^2, \quad F_r(0) = 1.$$

At level r , the only new function on the right is F_r^2 .

Proof. Condition on the split in (2), expand $(1+ab)^r = \sum_{k=0}^r \binom{r}{k} a^k b^k$, and use conditional independence. This gives (10). Coefficient extraction from the square of each F_k gives (11). $\square$

The cases $r = 1, 2$ lie in the mean/variance analysis of Disanto et al. [3]. More generally, the displayed identity is a one-line binomial expansion of their exact distributional recurrence. It therefore receives zero contribution credit and serves only as the interface for the strict higher-order theorem; no elementary solution is claimed at every level.

Theorem 4.2 (Strict higher-moment radius continuation). *There are numbers*

$$(12) \quad 1 = \rho_0 > \rho_1 > \rho_2 > \cdots > 0$$

such that ρ_r is the radius of convergence of F_r . For every $r \geq 1$,

$$(13) \quad F_r(z) = \frac{1}{\rho_r - z} + O(1) \quad (z \rightarrow \rho_r),$$

and

$$(14) \quad \limsup_{n \rightarrow \infty} (\mathbb{E} X_n^r)^{1/n} = \rho_r^{-1}.$$

The residual assertion begins at $r = 3$: the cases $r = 1, 2$ are retained only as the owned base of the continuation.

Proof. Proceed by induction from $F_0(z) = (1-z)^{-1}$ and $\rho_0 = 1$. For $r \geq 1$, separate the new unknown in (11):

$$(15) \quad F'_r = F_r^2 + G_r, \quad G_r = \sum_{k=0}^{r-1} \binom{r}{k} F_k^2.$$

On $|z| < \rho_{r-1}$, let U_r solve

$$(16) \quad U_r'' + G_r U_r = 0, \quad U_r(0) = 1, \quad U_r'(0) = -1.$$

The logarithmic derivative $-U_r'/U_r$ satisfies (15) and agrees with F_r near zero.

By induction, all terms in G_r except the one containing F_{r-1} are analytic at ρ_{r-1} , while (13) gives

$$(17) \quad G_r(x) = \frac{r}{(\rho_{r-1} - x)^2} + O\left(\frac{1}{\rho_{r-1} - x}\right) \quad (x \uparrow \rho_{r-1}).$$

Choose c with $1/4 < c < r$. Close enough to ρ_{r-1} , $G_r(x) \geq c(\rho_{r-1} - x)^{-2}$. Every nonzero solution of the Euler comparison equation

$$V'' + \frac{c}{(\rho_{r-1} - x)^2} V = 0$$

has infinitely many zeros accumulating at ρ_{r-1} : after putting $s = \rho_{r-1} - x$, it is a linear combination of $s^{1/2} \cos(\nu \log s)$ and $s^{1/2} \sin(\nu \log s)$, with $\nu = \sqrt{4c - 1}/2$. If U_r has not already vanished, choose two consecutive comparison zeros after the point where the coefficient inequality holds. The Sturm comparison theorem forces a zero of U_r strictly between them [12, Chapter 1]. Hence U_r vanishes strictly before ρ_{r-1} . Let ρ_r be its first positive zero. Then $0 < \rho_r < \rho_{r-1}$.

The coefficient G_r is analytic at ρ_r . A double zero of U_r would contradict uniqueness for (16), so the zero is simple. Consequently $-U_r'/U_r$ has the normalized local form (13).

It remains to identify the positive pole with the complex radius. The coefficients of F_r are positive. If its radius R were smaller than ρ_r , Pringsheim's theorem would force a singularity at the positive point R [4, Section IV.6]. Yet G_r is analytic there and U_r has no positive zero before ρ_r , so $-U_r'/U_r$ is analytic in a neighborhood of R , a contradiction. Thus $R = \rho_r$. Cauchy–Hadamard now gives (14); replacing the exponent $1/(n-1)$ by $1/n$ does not change the limit superior. $\square$

Remark 4.3 (Hard ceiling for $r \geq 3$). The local pole in (13) lies at the positive convergence radius, but the proof does not exclude other singularities of the same modulus. Accordingly, theorem 4.2 claims only (14) for $r \geq 3$. A statement such as $\mathbb{E}X_n^r \sim \rho_r^{-n}$ would require an additional dominant-singularity theorem and is not part of this paper. The stronger order-two asymptotic in Disanto et al. [3] is not weakened or reclaimed.

5. OWNED MEAN AND ENDPOINT NORMALIZATIONS

At $r = 1$, the forcing in (15) is explicit. This permits a complete coefficient asymptotic, unlike the higher levels.

Theorem 5.1 (Direct-owner mean, reproduced for normalization). *Let $M = F_1$ and $w = 1 - z$. Then*

$$(18) \quad M' = M^2 + \frac{1}{(1-z)^2}, \quad M(0) = 1,$$

and

$$(19) \quad M(z) = \frac{1}{w} \left[\frac{1}{2} - \frac{\sqrt{3}}{2} \tan \left(\frac{\sqrt{3}}{2} \log w - \frac{\pi}{6} \right) \right].$$

Its unique dominant singularity is the simple pole

$$(20) \quad \rho = 1 - \exp \left(-\frac{2\pi}{3\sqrt{3}} \right) = 0.7015639408 \dots$$

For some $R > \rho$,

$$(21) \quad \mathbb{E}X_n = \rho^{-n} + O(R^{-n}), \quad \rho^{-1} = 1.4253868276 \dots$$

In particular, the leading coefficient is one.

Proof. Equation (18) is the case $r = 1$ of proposition 4.1. Write $M = -U'/U$. Then

$$U'' + \frac{U}{(1-z)^2} = 0, \quad U(0) = 1, \quad U'(0) = -1.$$

With $\alpha = \sqrt{3}/2$, the solution is

$$(22) \quad U(z) = \frac{2}{\sqrt{3}} w^{1/2} \cos \left(\alpha \log w - \frac{\pi}{6} \right).$$

Since $M = U_w/U$, differentiation yields (19).

In $|z| < 1$, the number $w = 1 - z$ lies in the right half-plane, so the principal logarithm is analytic. The zeros of the cosine in (22) satisfy

$$w_k = \exp \left[\frac{2}{\sqrt{3}} \left(\frac{2\pi}{3} + k\pi \right) \right], \quad k \in \mathbb{Z}.$$

They are positive real because all complex zeros of cosine are real. The zero $k = -1$ gives (20). The zeros with $k \leq -2$ correspond to larger positive $z < 1$, while those with $k \geq 0$ give $|z| > 1$; the logarithm first branches at $z = 1$. Hence ρ is the unique singularity on its circle and is isolated from the next one. The logarithmic derivative of a simple zero has local form $1/(\rho - z) + O(1)$. Coefficient extraction gives (21). $\square$

The theorem is the shifted form of the Yule–Harding mean analysis in Disanto et al. [3, Section 5.2]; no part of it is counted as a residual advance.

The opposite edge of the distribution also has a closed form.

Proposition 5.2 (Fully owned caterpillar normalization). *For every $n \geq 1$, $X_n \geq n$. For $n \geq 2$,*

$$(23) \quad \min \text{supp } X_n = n, \quad \mathbb{P}(X_n = n) = \frac{2^{n-2}}{(n-1)!}.$$

Proof. For positive integers, $xy + 1 \geq x + y$, with equality exactly when $x = 1$ or $y = 1$. Induct on n in (2). If the root split has sizes i and $j = n - i$, then

$$1 + X_i X_j' \geq 1 + ij = i + j + (i-1)(j-1) \geq n.$$

Equality requires minimal values in both subtrees and an endpoint split $i = 1$ or $j = 1$. Iterating this condition gives precisely the planar combs. Among the $(n-1)!$ boundary orders, their number c_n satisfies $c_2 = 1$ and $c_n = 2c_{n-1}$ for $n \geq 3$. Thus $c_n = 2^{n-2}$, which proves (23). $\square$

The fact that the caterpillar minimizes the number of root configurations is already recorded by Disanto et al. [3, Section 2.4.4]. The displayed mass is exactly the Yule–Harding n -caterpillar probability printed by Chang and Fuchs [2, Table 1]; Rosenberg [10] is the earlier caterpillar-pattern owner. The proof is retained only as a translation to the adjacent-deletion encoding, and the entire proposition receives zero contribution credit.

6. EXACT COEFFICIENT CONTROLS AND PROOF DEPENDENCIES

The first marked polynomials make the coefficient interpretation concrete. They were obtained both from (9) and by averaging over every boundary order through $n = 9$.

TABLE 1. Exact expected cardinality-marked antichain polynomials. At $s = 1$, the owned root-configuration statistic satisfies $R_n = X_n - 1$. The machine-readable artifact continues through $n = 12$.

n	$a_n(s) = \mathbb{E}P_{T_n}(s)$	$\mathbb{E}X_n$
1	1	1
2	$1 + s$	2
3	$1 + 2s$	3
4	$1 + 3s + \frac{1}{2}s^2$	$\frac{13}{2}$
5	$1 + 4s + \frac{1}{2}s^2$	$\frac{37}{2}$
6	$1 + 5s + \frac{13}{5}s^2 + \frac{2}{15}s^3$	$\frac{131}{15}$
7	$1 + 6s + \frac{47}{10}s^2 + \frac{59}{90}s^3$	$\frac{556}{45}$
8	$1 + 7s + \frac{263}{35}s^2 + \frac{121}{63}s^3 + \frac{17}{315}s^4$	$\frac{787}{45}$

The standard-library verifier implements two independent finite calculations where such independence is literal: direct adjacent merging for each boundary permutation and dynamic programming from the split law. It also compares the marked tree polynomials with (6), raw moments with (11), the Euler-series logarithmic derivative with the mean coefficients, and the comb count with (23). The exact artifact `code/marked_antichain_coefficients.tsv` records $a_n(s)$, the first two moments, minimum mass, and support size through $n = 12$.

These checks do not prove the singularity statements. Conversely, the analytic proof does not turn the heap-labeling viewpoint into a second probability law: both temporal and structural calculations inherit the same uniform split from theorem 2.1. The honest dependency chain is

$$\text{uniform boundaries} \implies \text{BST split} \implies \begin{cases} \text{finite law and moment Riccati hierarchy,} \\ \text{heap-labeled tree and marked antichains.} \end{cases}$$

The two branches cross-check finite coefficients and extremal combs, while the Sturm–Pringsheim argument is specific to the temporal moment branch.

7. OWNERSHIP BOUNDARY AND CONCLUSION

Most importantly, Disanto et al. [3] own the same statistic under $R_n = X_n - 1$: the Yule split and finite law, root configurations and unmarked antichains, the mean Riccati equation and pole, and the second-moment/variance neighborhood all receive zero credit. Uniform random-BST splitting, Cartesian trees, forest ideals, Riccati linearization, Pringsheim’s theorem, and generic singularity extraction are also background. Andriantiana et al. [1] own fixed-tree cardinality-marked antichains, and Chang and Fuchs [2], Rosenberg [10] own the caterpillar probability neighborhood. The residual recorded here is only the Yule-averaged bivariate transform and, centrally, the strict pole/radius continuation for every $r \geq 3$. The all-order identity is a mechanical interface, not a separate advance. We do not claim that all equivalent terminology or unpublished work has been exhausted.

The product-plus-one encoding is useful because it turns the owned root-configuration statistic into a literal adjacent dynamics. Retaining a fixed-tree cardinality marker and averaging it under the Yule law yields the exact transform, while the triangular Riccati system lets the two low-order owner results propagate to a strict hierarchy at every higher order. Those higher moments stop at the exact limsup in remark 4.3. Novelty, priority, and external dissemination remain HOLD pending specialist review.

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